Causal Inference and Machine Learning

Heterogeneous Treatment Effects: Causal Trees, Forests, and Honest Estimation

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Quiz 1

What is the primary goal of estimating the Conditional Average Treatment Effect (CATE)?

1. To calculate the single average effect for the entire population.
2. To identify how treatment effects vary with observable characteristics X.
3. To replace randomized experiments with purely observational data.
4. To minimize the number of covariates used in a regression.

Quiz 1 Solution

1. To calculate the single average effect for the entire population.
2. To identify how treatment effects vary with observable characteristics X.
3. To replace randomized experiments with purely observational data.
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Conditional Average Treatment Effects (CATE)

- Standard ATE focuses on population averages; CATE ($\tau(x)$) estimates effects for specific subpopulations.
- Definition: $\tau(x) = \mathbb{E}[Y^{(1)} - Y^{(0)} \mid X = x]$.
- Objective: use machine learning to discover granular patterns without post-hoc subgroup selection.
- Key assumption: unconfoundedness, meaning treatment is as good as random conditional on covariates X.

Why Heterogeneity Matters

- Treatment effects can vary strongly across observable characteristics, so a single average effect can hide important structure.
- Heterogeneity is useful for understanding mechanism, targeting policies, and identifying who benefits most from an intervention.
- The challenge is that we need methods that preserve reproducibility and valid inference while still searching flexibly for patterns.

ML for Policy

- Machine learning helps build more granular statistical models and improves counterfactual prediction.
- It can also help control for confounding by using observables flexibly instead of relying on rigid functional forms.
- Causal inference needs valid confidence intervals and protection against false discovery, not only good prediction.

Quiz 2

What does "Honesty" signify in the context of Causal Trees?

1. The researcher must pre-register all hypotheses before looking at the data.
2. The algorithm uses different data for selecting splits and estimating leaf effects.
3. The tree must only split on covariates that are highly correlated with the outcome.
4. The treatment assignment must be perfectly balanced in every leaf.

Quiz 2 Solution

1. The researcher must pre-register all hypotheses before looking at the data.
2. The algorithm uses different data for selecting splits and estimating leaf effects.
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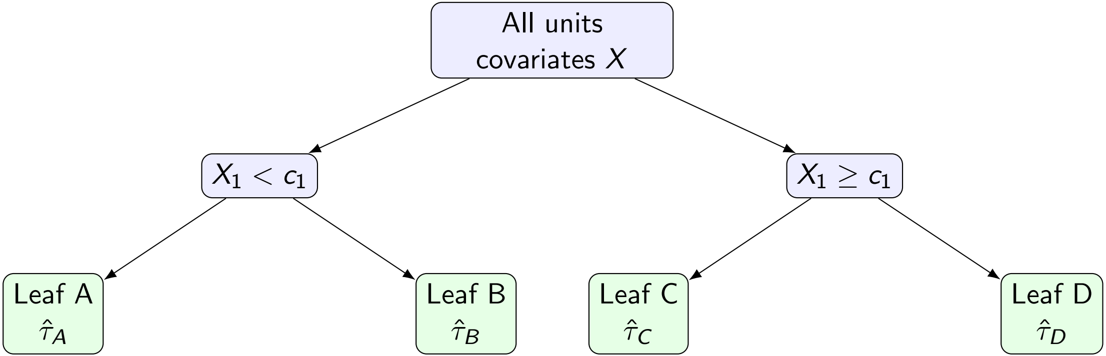
4. The treatment assignment must be perfectly balanced in every leaf.

Causal Trees: Recursive Partitioning

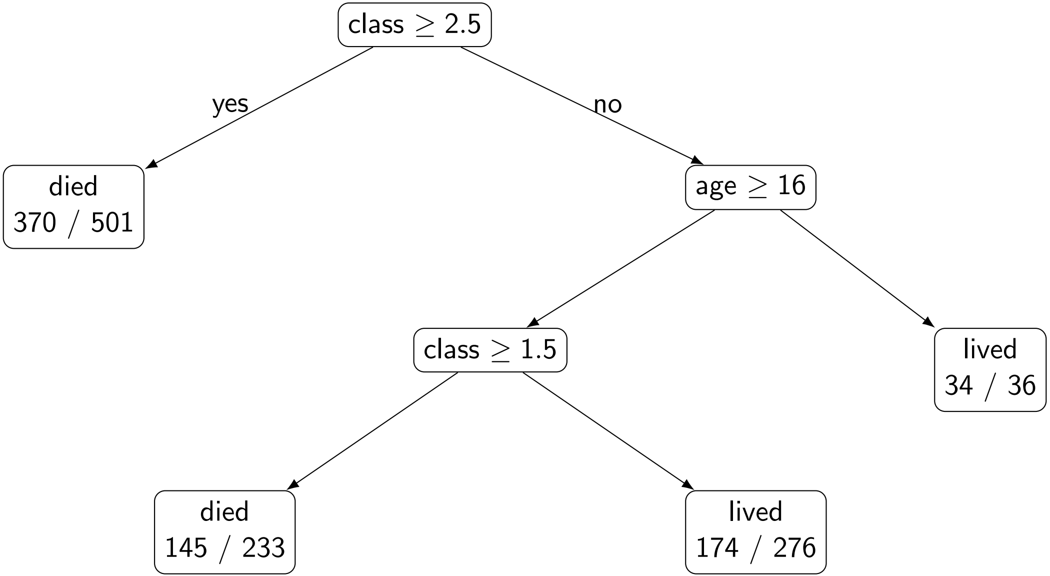
- Causal trees divide the covariate space into leaves to minimize mean squared error of treatment effects.
- The key difficulty is that individual treatment effects are unobserved, so the objective must be approximated indirectly.
- Trees are low-dimensional subgroup discovery with interpretable partitions.

Causal Tree Idea



- Each split tries to create more homogeneous treatment-effect regions.
- The tree is easy to explain, but different trees can be equally plausible when covariates are correlated.

Regression Tree Example (Titanic)



Honest Estimation

- The problem: using the same data to discover subgroups and estimate their effects produces bias and overfitting.
- The solution: split the sample into a training part for tree building and an estimation part for leaf effects.
- Honesty gives cleaner confidence intervals because the leaf estimates are not optimized on the same observations.

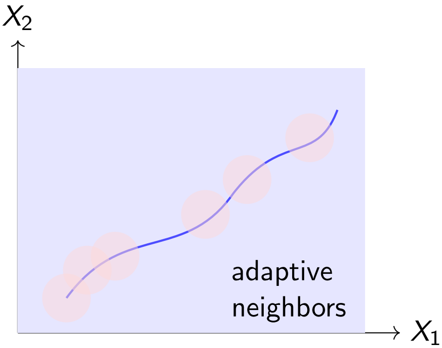
From Trees to Forests

- Causal forests average many honest trees to estimate $\tau(x)$ more smoothly.
- The forest view is close to adaptive nearest-neighbor matching, where nearby observations receive larger weights.
- Compared with a single tree, a forest reduces discontinuities at leaf boundaries.

Forest vs k-NN

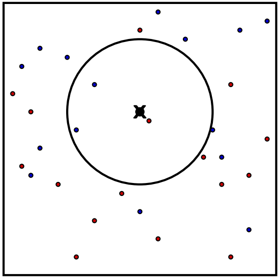
- Standard k-NN suffers from the curse of dimensionality because all neighbors become far away in high dimensions.
- Causal forests adaptively focus on covariates that matter for heterogeneity and downweight noise covariates.
- Asymptotic normality, which makes confidence intervals possible.

Causal Forest Intuition

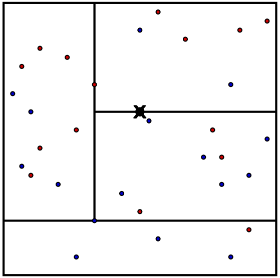


- Forests act like an adaptive neighborhood around the target point.
- This makes them more flexible than a single partition, especially when treatment effects vary smoothly.

k-NN vs Adaptive Partitioning



k-NN Neighborhood

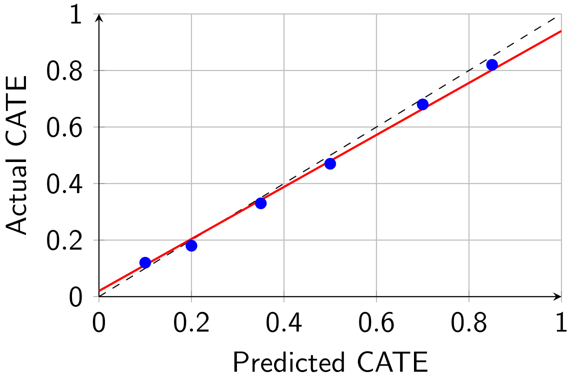


Adaptive Partitioning

Calibration and Heterogeneity

- Use calibration checks to see whether higher predicted CATEs correspond to larger actual effects.
- A best linear predictor on held-out data is a simple diagnostic: a slope near one suggests good calibration.
- Quartile plots help reveal whether the model really orders parameters by treatment benefit.

Calibration Plot



- The 45-degree line is the ideal target.
- Departures from that line show under- or over-confidence in the estimated treatment effects.

Quiz 3

Why might one use a Local Linear Forest instead of a standard Causal Forest?

1. To reduce the computational time required for training.
2. To better handle smooth, non-step-like relationships in the data.
3. Because standard forests cannot handle more than two covariates.
4. To avoid the need for sample splitting.

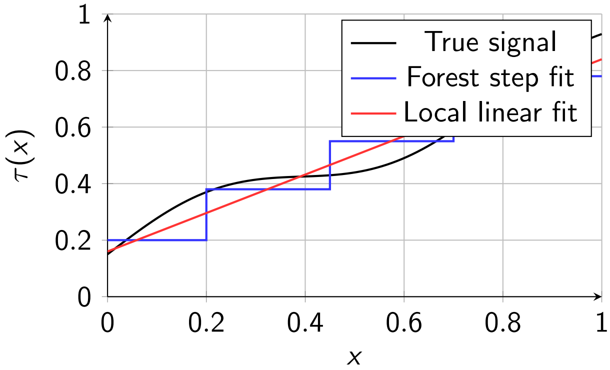
Quiz 3 Solution

1. To reduce the computational time required for training.
2. To better handle smooth, non-step-like relationships in the data.
3. Because standard forests cannot handle more than two covariates.
4. To avoid the need for sample splitting.

Local Linear Forests

- Standard forests often fit step functions, which can be inefficient for smooth relationships.
- Local linear forests add local slope correction near boundaries to improve fit.
- This is especially useful in economic data where treatment effects vary smoothly with covariates.

Step vs Smooth Fit



- The local linear correction helps recover smooth trends that a step function flattens.

Locally Linear Forest

Locally linear regression with ridge penalty:

$$\begin{pmatrix} \hat{\mu}(x) \\ \hat{\theta}(x) \end{pmatrix} = \arg\min_{\mu,\theta} \sum_{i=1}^{n} \alpha_i(x)(Y_i - \mu(x) - (X_i - x)\theta(x))^2 + \lambda\|\theta(x)\|_2^2$$

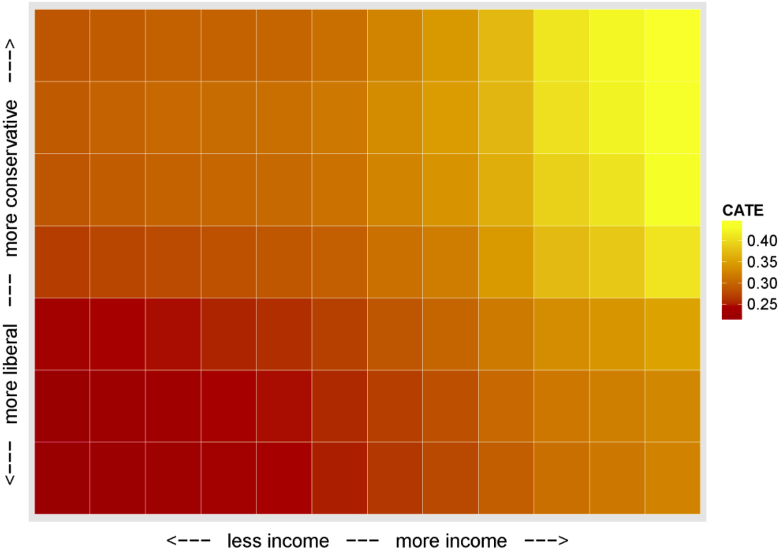
In matrix form:

$$\begin{pmatrix} \hat{\mu}(x) \\ \hat{\theta}(x) \end{pmatrix} = (X^T AX + \lambda J)^{-1} X^T AY$$

Application: GSS Experiment

- GSS: a survey experiment where respondents were randomly asked about welfare or assistance to the poor.
- The causal tree found heterogeneity linked to income, education, and political views.
- The key teaching point is that all groups showed positive treatment effects, but conservative respondents reacted more strongly.

Application: GSS Experiment



Summary of CATE Estimation

- Causal trees offer interpretable subgroup discovery with honest inference.
- Causal forests provide flexible, high-dimensional estimation by averaging honest trees.
- Calibration is essential when signal is weak, because apparent heterogeneity can be mostly noise.